

	A	B	C	D	E	F	G	H
1	Power Budget Example							
2	Team Number:	209						
3	Project Name:	Door Sensor						
4	Team Member Names:	Bryce, Matthew, Dylan						
5	Version:	0.01 Pre Release						
6								
7	A. List ALL major components (active devices, integrated circuits, etc.) except for power sources, voltage regulators, resistors, capacitors, or passive elements							
8	All Major Components	Component Name	Part Number	SupplyVoltageRange	#	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
9		Op Amp	MCP6004-I/P	+1.6 - 6V	1	0.17	0.17	mA
10		Relay Switch	CST10T2CR	+0.1 - 20V	1	20	20	mA
11		Dynamic Speaker Unit	SP-4005Y	+2.8 - 3.4V	1	42	42	mA
12		5V regulator	AP7375-50SA-7	+5V - 45V	1	1500	1500	mA
13	m							
14								
15	B. Assign each major component above to ONE power rail below. Try to minimize the number of different power rails in the design.							
16	+5V Power Rail	Component Name	Part Number	SupplyVoltageRange	#	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
17		PSoC™ 4 BLE Module	PIC18F57Q10	+1.8 - 5.5	1	200	200	mA
18		Op Amp	MCP6004-I/P	+1.6 - 6V	1	0.17	0.17	mA
19		Relay Switch	CST10T2CR	+0.1 - 20V	1	20	20	mA
20		Dynamic Speaker Unit	SP-4005Y	+2.8 - 3.4V	1	42	42	mA
21						Subtotal	262.17	mA
22						Safety Margin	25%	
23						Total Current Required on +5V Rail	327.7125	mA
24								
25	c2. Regulator or Source Choice	5V regulator	AP7375-50SA-7	+5V - 45V	1	1500	1500	mA
26						Total Remaining Current Available on +5V Rail	1172.2875	mA
27	C. For each power rail above, select a specific voltage regulator using the same process as for major component selection. Confirm that the Total Remaining Current Available on each							
28								
29	D. Select a specific external power source (wall supply or battery) for your system, and confirm that it can supply all of the regulators for all of the power rails simultaneously. If you need							
30	External Power Source 1	Component Name	Part Number	SupplyVoltageRange	Output	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
31	Power Source 1 Selection	Plug-in Wall Supply	N/A	110VAC	+24V	15000	15000	mA
32								
33	Power Rails Connected to External Power Source 1							
34		5V regulator	AP7375-50SA-7	+5V - 45V	1	1500	1500	mA
35								
36						Total Remaining Current Available on External Power Source 1	13500	mA
37								
38	External Power Source 2	Component Name	Part Number	SupplyVoltageRange	Output	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
39	Power Source 2 Selection	Battery	AA	+120V	1.5V	25000	25000	mA
40								
41	Power Rails Connected to External Power Source 2	5V regulator	AP7375-50SA-7	+5V - 45V	1	1500	1500	mA
42								
43								
44						Total Remaining Current Available on External Power Source 2	23500	mA
45								
46	E. Calculate Battery Life (if applicable). For each battery, also check the worst-case lifetime of the battery by indicating the capacity in mAh.							

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47	Component Name		Part Number	SupplyVoltageRange	Capacity(mAh)		RequiredByRegulators	
48		Battery	AA	1.5V		500	1500	
49						Battery Life	0.333333333	hours
50	Notes							
51	External Supply Voltage should be determined by the dropout voltage for highest-voltage regulator (e.g., +14V for a +12V regulator).							
52	If you have multiple units in your design (e.g., a base unit and remote unit) then you need a separate power budget for each unit							